

Taewoon Kang

Integrated M.S. and Ph.D. Student

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EDUCATION

Mar. 2023 ~ Present	Korea University Department of Computer Science and Engineering <i>Advisor: Gunjae Koo</i> <i>Ph.D. Student</i> GPA: 4.35 / 4.5	Seoul, Korea
Mar. 2019 ~ Aug. 2022	Sangmyung University Department of System Semiconductor Engineering Thesis: An Implementation of CNN-based Real-Time Face Mask Detector using Jetson Nano <i>Advisor: Yongwoo Kim</i> <i>B.S in System Semiconductor Engineering</i> GPA: 4.28 / 4.5	Cheonan, Korea

RESEARCH INTERESTS

- AI Accelerator
- HW/SW Co-Design SoC
- Processing for AI (NPU)
- Computer architecture
- Processing-in memory

EXPERIENCES

- Visiting Scholar at Ming Hsieh Department of Electrical and Computer Engineering, University of Southern California, United States (Jan. 2026 ~ Present) / Visiting Scholar under the supervision of Prof. Murali Annavam
- Intern at SK Telecom, Korea (Jul. 2023 ~ Aug. 2023) / Develop spam detection methods using AI

PUBLICATIONS (INTERNATIONAL)

1. (Under review) Narae Kim, Wonjae Choi, Taewoon Kang, Yujin Kim, Seongpil Yang, Myung Kuk

Yoon, Yunho Oh, Gunjae Koo, "Fine-Grained DRAM Refresh Exploiting Divergent Bit-Error Tolerance in AI Workloads", *DESIGN, AUTOMATION, AND TEST IN EUROPE (2027)*

2. (Under review) Taewoon Kang, Jonghyun Lee, Yujin Kim, Murali Annavaram, Gunjae Koo, "LogPIM: Low-Overhead Logarithmic Arithmetic for HBM-Based Processing-in-Memory", *IEEE COMPUTER ARCHITECTURE LETTERS*
3. (Under review) Jong Hyun Jeong, Yuan Feng, Taewoon Kang, Gunjae Koo, Hyeran Jeon, "GPO: Orchestrating GPU-PIM Computing for Sparse-Dense Workload Acceleration.", *Proceedings of the 33rd IEEE International Symposium on High-Performance Computer Architecture (2027)*
4. (To appear) Jonghyun Lee, Taewoon Kang, Pratheek B, Yongqin Wang, Rachit Rajat, Mengyuan Li, Gunjae Koo, Daniel Wong, Murali Annavaram, "FLASH: Fault-Led, Attested, Secure Hybrid-Key Checkpoint for Confidential GPU.", *Proceedings of the 2026 ACM Annual Technical Conference (2026)*
5. Taewoon Kang, Namhun Kim, Jihun Lee, Gunjae Koo, "SoftmaxPIM: An HBM-Based PIM Architecture for Accelerating GeMV-Softmax Execution Pipeline", *IEEE COMPUTER ARCHITECTURE LETTERS*, (2026)
6. Taewoon Kang, Namhun Kim, Geonwoo Choi, Taeweon Suh, Gunjae Koo, "SparsePIM+: Accelerating SpMV on HBM-Based PIM via Logic-Die Accumulators with Opportunistic TSV Utilization", *JOURNAL OF SYSTEMS ARCHITECTURE*, (2026)
7. Sunchae Kim, Taewoon Kang, Sangwon Shin, Taeweon Suh, Yibin Yang, Gunjae Koo, "SumcheckPIM: An Efficient HBM-Based PIM Architecture for Linear Complexity Zero Knowledge Proofs", *ACM INTERNATIONAL CONFERENCE ON SUPERCOMPUTING*, (2026)
8. Taewoon Kang, Geonwoo Choi, Taeweon Suh, Gunjae Koo, "SparsePIM: An Efficient HBM-Based PIM Architecture for Sparse Matrix-Vector Multiplications", *ICS '25: PROCEEDINGS OF THE 39TH ACM INTERNATIONAL CONFERENCE ON SUPERCOMPUTING*, (2025)

PUBLICATIONS (DOMESTIC)

1. Seongpil Yang, Taewoon Kang, "Performance Analysis of a Per-Row Activation Counting Mechanism in DRAM", *The Institute of Electronics and Information Engineers*, 2025 summer conference, (2025)
2. Sang-un Jo, Tae-woon Kang, Yongwoo Kim, "A Design and Implementation of MP3 Player using ARM Cortex-M0 DesignStart", *The Korean Institute of Electrical Engineers Workshop*, Yeosu, Korea (Jul. 2022)
3. Sang-un Jo, Tae-Woon Kang, Yongwoo Kim, "A design and verification of ARM Cortex-M0 DesignStart MCU system using open source IP", *ICS 2022*, Jeonju, Korea (Apr. 2022)
4. Taewoon Kang, Yongwoo Kim, "An Implementation of CNN-based Real-Time Face Mask Detector using Jetson Nano ", *JKAIS*, (Vol. 23, 2022)

PATENTS (DOMESTIC)

1. 구건재, 강태운, "메모리 내 직접 연산 기반 희소 행렬-벡터 곱 연산 장치 및 방법", KR-Registration No. 10-3008421

AWARDS AND HONORS

- The top prize of CISC Career day, Sangmyung University, Korea (May. 2022)
- Header (in one's dept.) Scholarship, Sangmyung University, Korea (Mar. 2022)
- Header (in one's dept.) Scholarship, Sangmyung University, Korea (Sep. 2021)
- "BONGSA" Scholarship (Leader of Department), Sangmyung University, Korea (Sep. 2021)

- Secondary Header (in one's dept.) Scholarship, Sangmyung University, Korea (Mar. 2021)
- Work Scholarship, Sangmyung University, Korea (Mar. 2021)
- "BONGSA" Scholarship (Leader of Department), Sangmyung University, Korea (Mar. 2021)
- Secondary Header (in one's dept.) Scholarship, Sangmyung University, Korea (Sep. 2020)

COURSE WORK

- Parallel processing, AI accelerator architecture
- Advanced computer architecture, Storage system architecture
- ARM Cortex-M series IP implementation
- Electrical circuit, Microelectronic circuit
- Semi-custom ASIC design with Synopsys design tool, VLSI design
- Digital logic circuit, Computer architecture and design, Design with Verilog, SoC design
- Semiconductor Physics, Semiconductor fabrication process
- Calculus, Engineering mathematics

SKILLS AND TECHNIQUES

- C, C++, Python
- GEM5, DRAMsim3
- Chip design verification with Xilinx/Altera FPGA
- Python (Pytorch)/ C language (Atmel, Keil)
- Verilog 2001 / SystemVerilog
- ARM Cortex-M0/M3 implementation
- RISC-V Simple processor design
- Semi-custom ASIC Design with Synopsys EDA tool

TEACHING EXPERIENCE

- Teaching Assistant
 - Basic of software: Spring 2025
 - Digital logic design: Spring 2023, Fall 2023

ACADEMIC SERVICE

- Artifact Evaluation Committee
 - IEEE/ACM International Symposium on Microarchitecture (MICRO); 2026